

Discharge Copilot

High-Level & Low-Level Design

Legacy system (MediTrack HMS v3.2) · Streamlit POC · Integrated pilot

Document	HLD & LLD: legacy as-is, POC, integrated system
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Prepared by	Forward Deployed Engineering team (FDE + AI Engineer)
Reviewed with	Hospital IT (S. Pawar), Medical Superintendent, TPA desk, Cardiology (Dr. A. Mehta)
Code	Branches part-1-legacy → part-2-poc → part-3-integrated
Diagram legend	Legacy / MediTrack Copilot (new) AI / LLM External service People Deterministic data

Note: CityCare Hospitals, all patients, doctors, TPAs and figures are fictional and exist for teaching. In the story MediTrack runs on Oracle; the demo implementation uses SQLite with the same schema.

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1. Introduction

1.1 Purpose & scope

This document describes the design of the **CityCare Discharge Copilot**, an AI-assisted discharge-summary system added to CityCare Hospitals' 2011-era Hospital Management System, **MediTrack HMS v3.2**. It covers three stages:

Stage	What it is	Where in this document
Legacy (as-is)	MediTrack as it exists: architecture, data model, process, constraints and hidden extension points	Section 3
POC	A standalone Streamlit application that proves the AI can draft summaries safely on de-identified historic data	Sections 4 (HLD) and 5 (LLD)
Integrated	The pilot system that runs next to MediTrack, reads its replica, appears as a MediTrack tab and writes back signed PDFs	Sections 6 (HLD) and 7 (LLD)

1.2 Audience

Hospital IT, clinical leadership, compliance, and the engineering team. Sections 2, 4 and 6 are readable by non-engineers; Sections 3.4–3.6, 5 and 7 are for engineers.

1.3 Glossary

Term	Meaning
IPD / OPD	In-patient (admitted to a bed) / out-patient department
UHID · IP No	Lifelong hospital patient ID · ID of one admission
Consultant · RMO	Senior treating doctor · Resident Medical Officer (junior doctor who writes most summaries)
ICD-10 · CPT	Diagnosis codes (e.g. I21.0 anterior STEMI) · procedure codes (e.g. 92928 coronary stent)
Provisional / final diagnosis	Working diagnosis at admission / confirmed diagnosis at discharge; insurers compare them
LIS · HL7 v2	Lab Information System · message standard used by the lab to send results
Cash · Cashless	Patient pays directly · insurer pays the hospital, requiring pre-authorization and final approval
TPA	Third-Party Administrator: processes cashless claims on behalf of the insurer
Pre-auth · Final approval · Query	Approval at admission · approval of the final bill at discharge · insurer pushback asking for missing information
Implant sticker	Label from a stent / joint package (brand, size, lot); insurers require it for implant procedures
Discharge TAT	Turnaround time from "discharge advised" to "patient leaves"
MRD · MIS	Medical Records Department · Management Information System (reporting)
ARPOB	Average Revenue Per Occupied Bed per day
NABH · DPDP Act	Indian hospital accreditation body · Digital Personal Data Protection Act, 2023

2. Business context

2.1 The in-patient discharge cycle

Every admitted patient passes through the cycle below. For **cashless** (insured) patients, the hospital cannot release the patient until the insurer, via the TPA, approves the final bill. That approval requires a complete discharge summary and supporting documents.

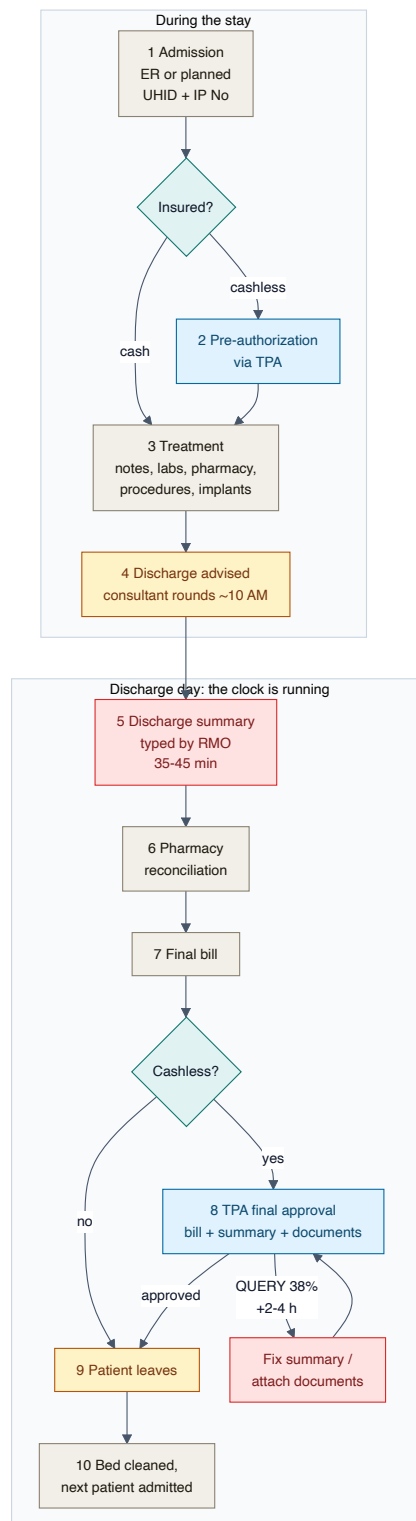


Figure 1. In-patient discharge cycle. Red steps are where the time is lost.

2.2 Problem statement

Stated problem (CEO): "Doctors are drowning in paperwork. Build us a ChatGPT for discharge summaries."

Real problem (from data): Insured patients wait **~6.3 hours** to go home after discharge is advised (vs ~2.5 h for cash patients) because discharge summaries are **slow to write** and often **incomplete for insurers**: 38% of TPA submissions come back with a query. This blocks beds and is the top patient complaint.

2.5 h

Cash discharge TAT

6.3 h

Cashless discharge TAT

38.1%

TPA submissions queried

35-45 min

RMO time per summary

2.3 Goals & success criteria (pilot, Cardiology Ward 4B, 4 weeks)

Goal	Metric	Baseline	Pilot target
Faster summaries	Discharge advised → draft ready	35-45 min of typing	< 1 minute
Fewer insurer queries	TPA query rate	38%	< 15%
Faster discharge	Cashless discharge TAT	6.3 h	< 3.5 h
Clinical safety	Hallucinated drugs / tests in drafts	n/a	0 (and 100% doctor sign-off)
Adoption	Share of pilot discharges signed via Copilot	0%	> 80%
Zero disruption	Changes to MediTrack code / DB writes	n/a	None (configuration only)

2.4 Stakeholders

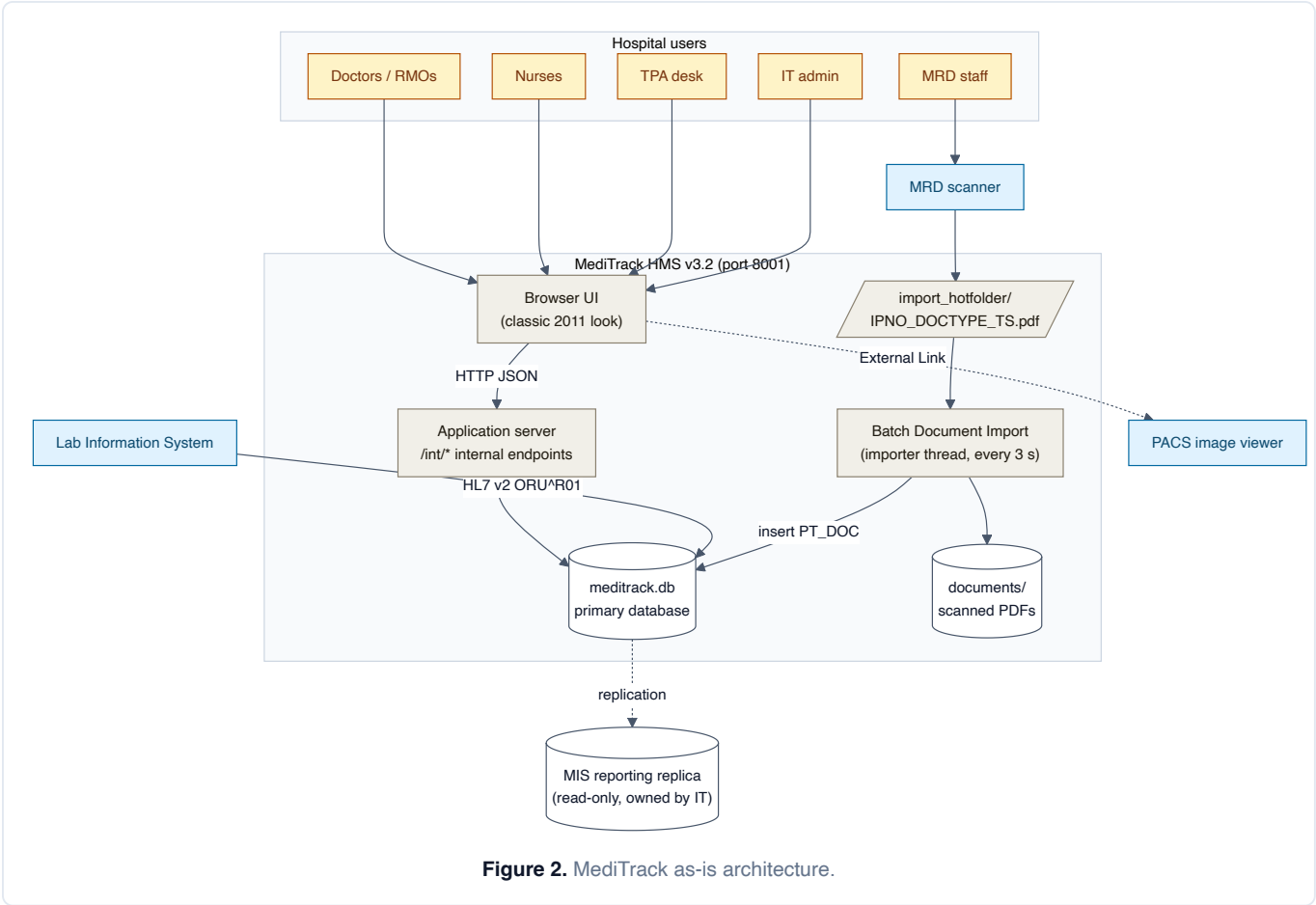
Stakeholder	Interest	Touchpoint
Consultants, RMOs	Less clerical work; clinical accuracy; signing responsibility	AI Discharge Draft tab in MediTrack
TPA desk	Complete first-time submissions	TPA-readiness checklist
Hospital IT (Suresh)	No risk to MediTrack; security	Replica, External Links config, hot-folder
CEO / CFO	Bed turnover, revenue, complaints	Copilot Monitor
Compliance	DPDP Act, NABH documentation, audit trail	PII minimisation, audit events, Logfire

3. Legacy system: MediTrack HMS v3.2 (as-is)

3.1 Overview & constraints

Attribute	Value
Product	MediTrack HMS v3.2.1 (Medisoft Solutions Pvt. Ltd., 2011). Vendor no longer trading; no source support.
Demo implementation	FastAPI application server + SQLite (meditrack.db) + React UI styled as 2011; port 8001
Production (story)	Java monolith on Oracle, 400+ tables, read-only MIS reporting replica maintained by IT
Knowledge holder	S. Pawar ("Suresh"), Hospital IT: the only person who understands the schema
APIs	/int/* endpoints exist only for MediTrack's own UI : undocumented, no third-party auth; IT policy forbids external use
Integrations	LIS → lab results over HL7 v2; MRD scanner → batch document import hot-folder; PACS viewer via External Link
Hard constraints	No code changes; no writes to the primary database; no downtime; users (300+ doctors and nurses) must not be retrained on a new application

3.2 As-is architecture



3.3 As-is discharge process

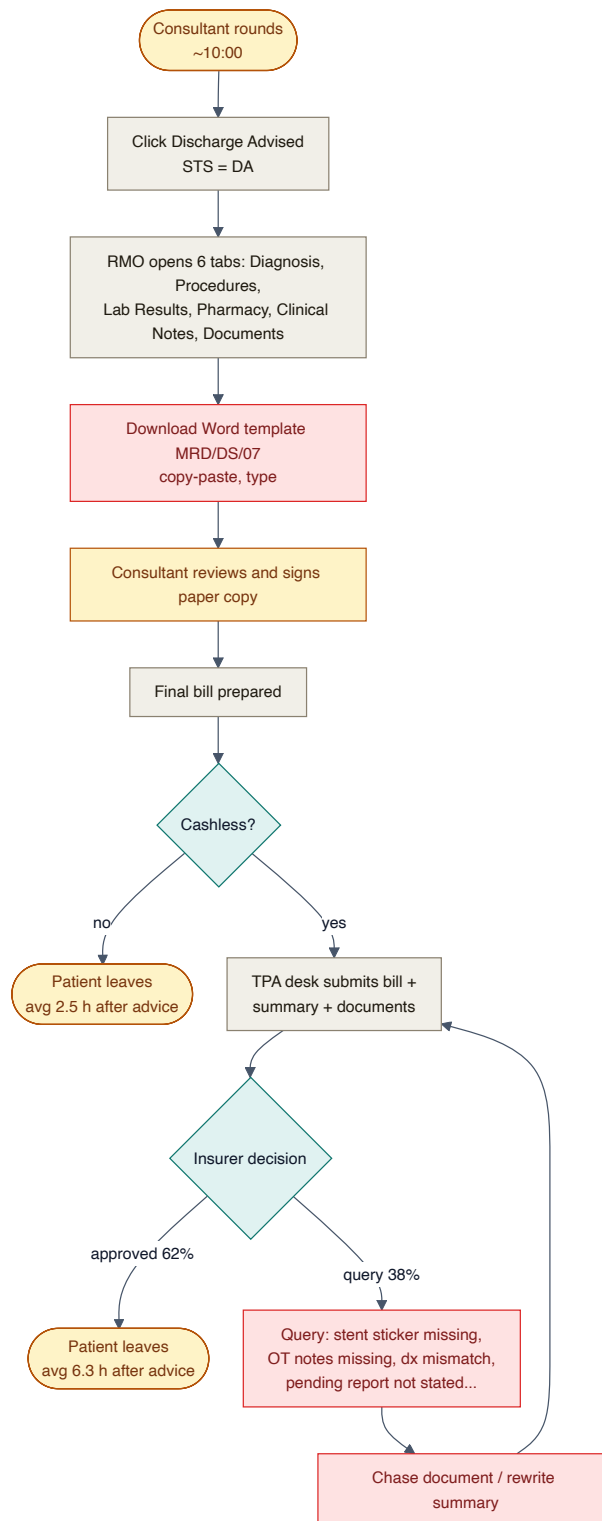


Figure 3. As-is discharge process in MediTrack, with the delay loop.

3.4 Data model

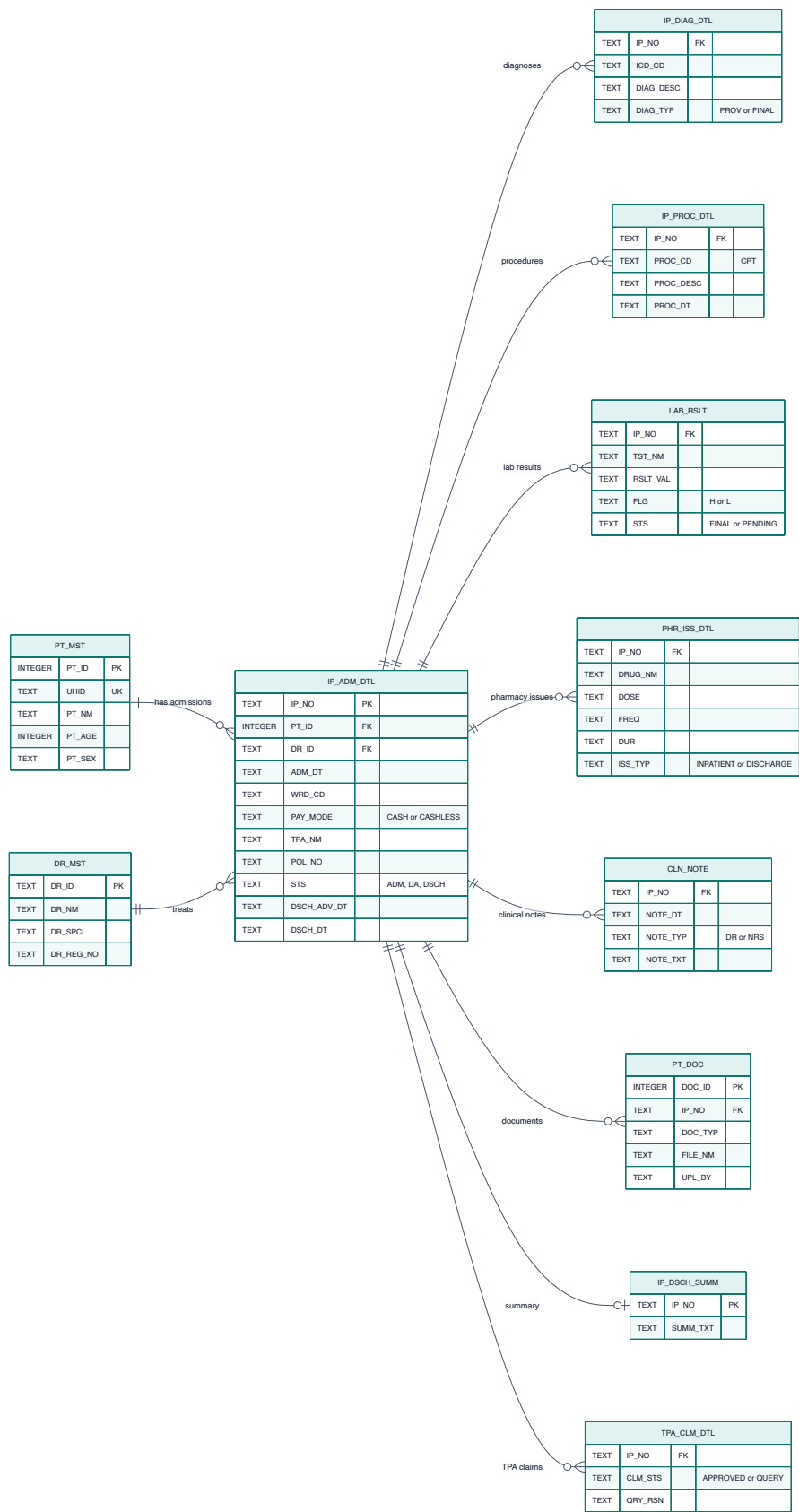


Figure 4. MediTrack data model (clinical core). System tables `SYS_EXT_LNK` and `SYS_AUDIT` are described in 3.5.

3.5 Table dictionary

Table	Holds	Key columns / notes
PT_MST	Patient master	UHID unique; name, age, sex, mobile, address
DR_MST	Doctor master	DR_REG_NO (medical council registration) is required by insurers
IP_ADM_DTL	One row per admission	STS lifecycle ADM → DA → DSCH ; DSCH_ADV_DT starts the discharge clock
IP_DIAG_DTL	Diagnoses	DIAG_TYP PROV vs FINAL. A FINAL diagnosis is required for TPA approval.
IP_PROC_DTL	Procedures	CPT code drives the required documents (see 5.6)
LAB_RSLT	Lab results fed by LIS (HL7)	STS PENDING = sent out, not yet reported
PHR_ISS_DTL	Pharmacy issues	ISS_TYP DISCHARGE = the discharge prescription (source of truth for meds)
CLN_NOTE	Free-text doctor and nursing notes	The main input the LLM summarises
PT_DOC	Scanned / uploaded documents	DOC_TYP: CATH_RPT, IMPLANT_STKR, OT_NOTE, CONSENT, ID_PROOF, DSCHSUMM...; UPL_BY BATCH_IMPORT for hot-folder imports
IP_DSCH_SUMM	Manually typed summaries	Historic ground truth for POC evaluation
TPA_CLM_DTL	Insurer final-approval submissions	CLM_STS APPROVED / QUERY with QRY_RSN; source of the 38% baseline
SYS_EXT_LNK	Admin-configurable External Links (v3.1, 2016)	LNK_URL with {IP_NO} / {UHID} placeholders; DISP_MD LINK (new window) or TAB (embedded patient-file tab)
SYS_AUDIT	Audit trail	User, action, reference

3.6 Internal endpoints (MediTrack UI only, not for integration)

Method	Path	Purpose
GET	/int/census	Current in-patients (STS ADM, DA)
GET	/int/discharged	Last 60 discharged patients
GET	/int/ip/{ip_no}	Full patient file: admission, diagnoses, procedures, labs, meds, notes, docs, summary, claims, active links
POST	/int/ip/{ip_no}/discharge-advise	ADM → DA, stamps DSCH_ADV_DT
POST	/int/ip/{ip_no}/discharge	DA → DSCH (final discharge)
POST	/int/ip/{ip_no}/manual-summary	Save typed summary
GET	/int/ip/{ip_no}/summary-template	Download Word template (.doc)
POST · GET	/int/ip/{ip_no}/docs · /int/docs/{doc_id}	Upload / view documents
GET · POST	/int/admin/extlinks[{id}/toggle]	List / add / enable-disable External Links
GET	/int/mis/discharge-tat · /int/audit	MIS discharge TAT report · audit trail

3.7 Extension points (discovered with Hospital IT)

Extension point	Originally built for	How the Copilot uses it
MIS reporting replica	Monthly management reports	Read: all patient data is read from the replica, never from the primary DB
External Links, display = TAB	PACS image viewer (2016)	Entry point: one configuration row adds the "AI Discharge Draft" tab to every patient file
Batch document import hot-folder	MRD scanner (2013)	Write-back: the signed summary PDF is dropped here and filed by MediTrack itself

3.8 Baseline metrics (from the MIS replica, last quarter)

Measure	Value	Source
Discharges analysed	151 (67 cash, 84 cashless)	IP_ADM_DTL where STS = DSCH
Avg discharge TAT: cash / cashless	2.5 h / 6.3 h	DSCH_DT – DSCH_ADV_DT
TPA query rate	38.1% (32 of 84)	TPA_CLM_DTL
Top query reason	Stent implant sticker / invoice not attached (9)	QRY_RSN

4. POC: High-Level Design

4.1 Objectives & non-goals

Objectives

- Prove an LLM can draft discharge-summary narrative **without inventing facts**
- Measure quality against doctors' own summaries (evals)
- Prove latency and cost are acceptable
- Get clinician and TPA-desk feedback (business verification)
- Validate the TPA-readiness rules against real query reasons

Non-goals

- No connection to MediTrack or any live hospital system
- No identifiable patient data (de-identified export only)
- No signing, write-back, or production hardening
- No UI polish; Streamlit is deliberately throwaway

4.2 Architecture

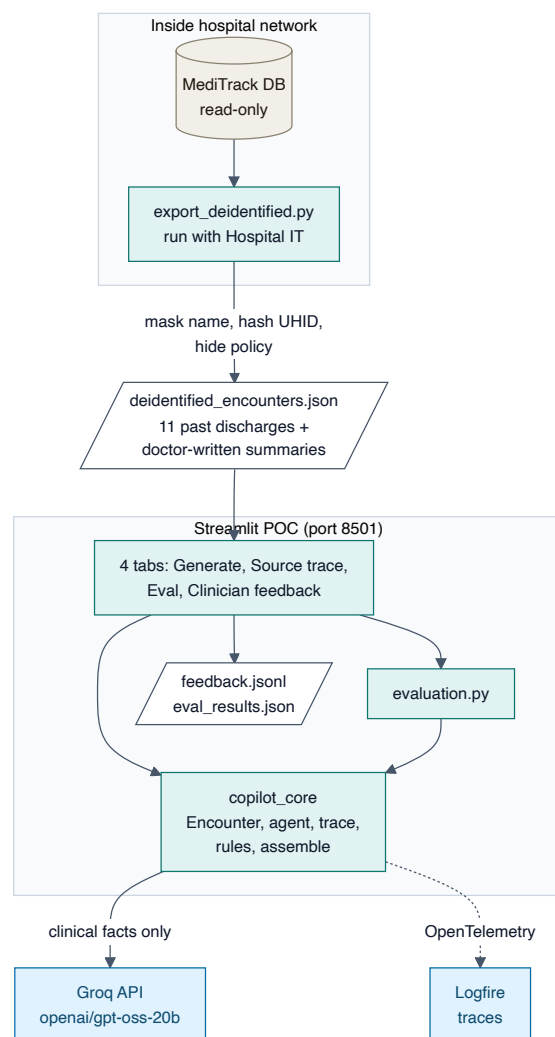


Figure 5. POC context and architecture. The only contact with MediTrack is a one-off, read-only, de-identified export.

4.3 What the AI writes vs what is deterministic

Summary section	Produced by	Rationale
Final diagnosis + ICD-10	MediTrack data	Legal and insurance fact; already coded
Discharge medications	Pharmacy data	A hallucinated drug or dose is a patient-safety event
Reports pending at discharge	Lab data	Frequent TPA query when omitted
TPA-readiness checklist	Rules engine	Insurer rules are known, so code them
Presenting complaints, hospital course, procedure findings, key investigations, condition at discharge, follow-up, diet & activity	LLM draft	Turning scattered free-text notes into a coherent narrative is where LLMs add value. Always doctor-editable.

Design principle: the best AI solution here is roughly 30% AI and 70% deterministic engineering. The LLM is used only where no source of truth exists.

4.4 Technology stack

Concern	Choice	Why
UI	Streamlit 1.64	Fast to build, disposable, familiar to data teams
Agent framework	Pydantic AI 2.50	Typed, validated structured output; native Logfire instrumentation
Model	groq:openai/gpt-oss-20b	Open-weight, very fast on Groq, low cost
Structured output	NativeOutput(SummaryDraft)	Provider-enforced JSON schema (Groq rejects gpt-oss tool-call output)
Observability	Logfire (OpenTelemetry)	Prompt, output, tokens, latency per run
Data	JSON export; pandas for eval tables	No database needed

4.5 Evaluation strategy

Each of the 11 historic encounters is drafted by the AI and scored automatically against the patient's source data and the doctor's original summary. Clinicians and the TPA desk then score a sample by hand (Clinician feedback tab). Automatic metrics are defined in 5.7.

5. POC: Low-Level Design

5.1 Modules

Module	Responsibility
copilot_core/models.py	Pydantic contracts: Encounter (input), SummaryDraft (LLM output), DraftResult (run metadata)
copilot_core/encounter.py	The only module that knows MediTrack's schema: SQL → Encounter ; deidentify()
copilot_core/agent.py	Pydantic AI agent, prompt, model settings, retry, fallback chain, cache
copilot_core/trace.py	Source-trace guardrail and highlighting
copilot_core/rules.py	TPA-readiness checklist and score
copilot_core/assemble.py	Final summary = deterministic facts + AI narrative
copilot_core/observability.py · config.py	Logfire setup (only if token present) · environment configuration
poc/export_deidentified.py	Read-only export of past discharges with summaries
poc/evaluation.py · poc/app.py	Eval metrics · Streamlit UI
scripts/warm_cache.py	Pre-generates live drafts for every patient (offline fallback)

5.2 Data contracts

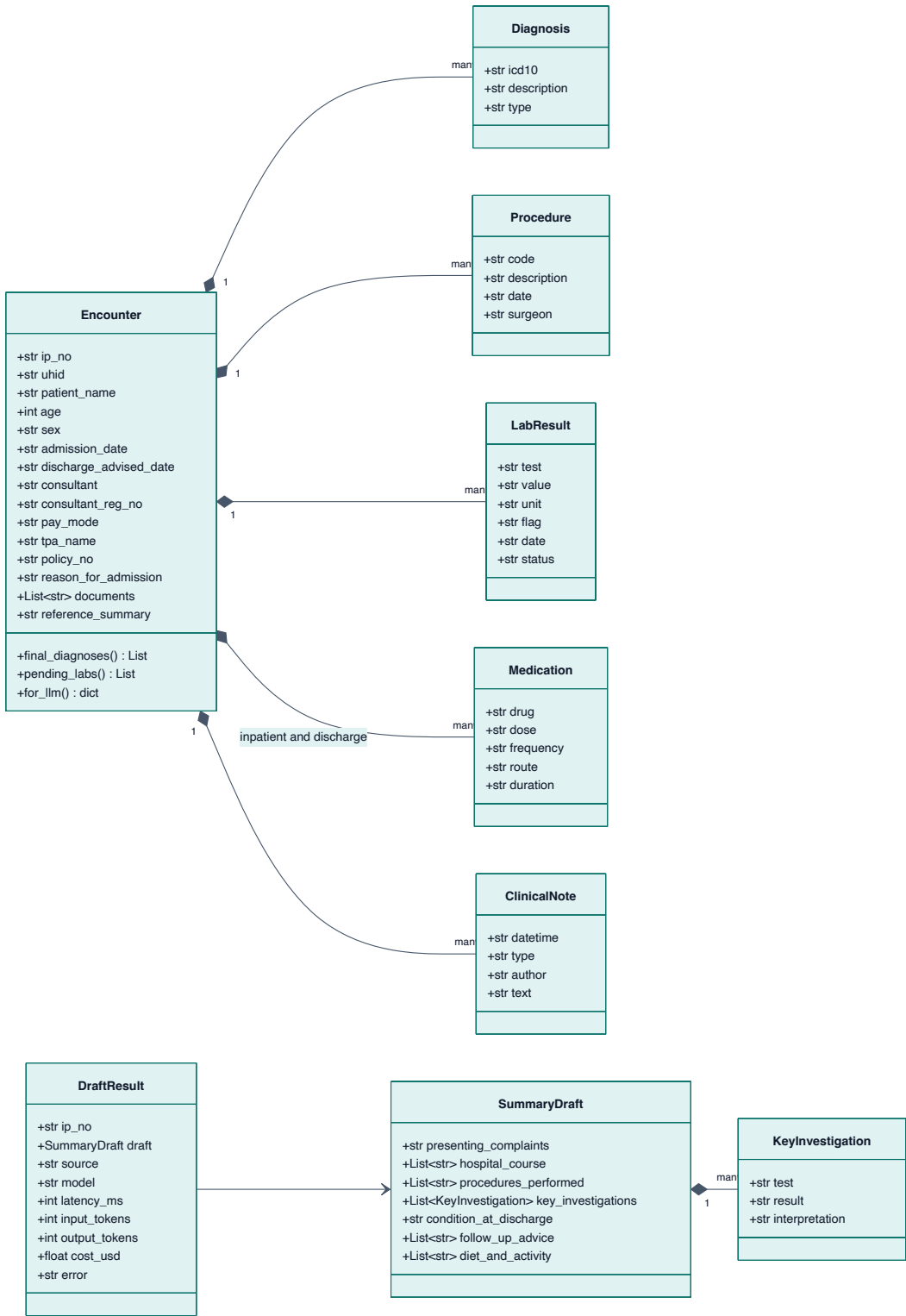


Figure 6. Data contracts. `for_llm()` excludes `ip_no`, `uhid`, `patient_name`, `policy_no`, `tpa_name`, `consultant_reg_no`, `documents`, `discharge_meds` and `reference_summary`.

5.3 Draft generation & fallback chain

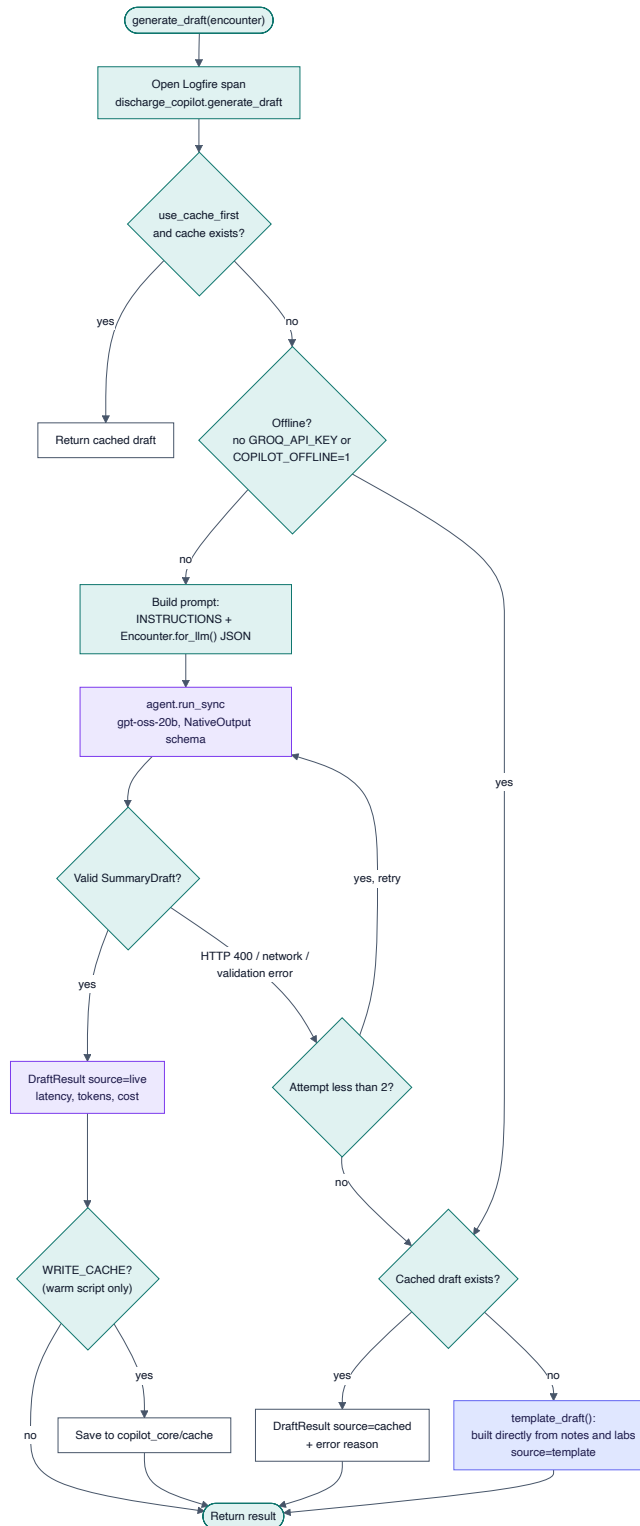


Figure 7. Draft generation with the three-level fallback: live LLM → cached draft → template. A demo or a ward never gets an empty screen.

5.4 Agent configuration

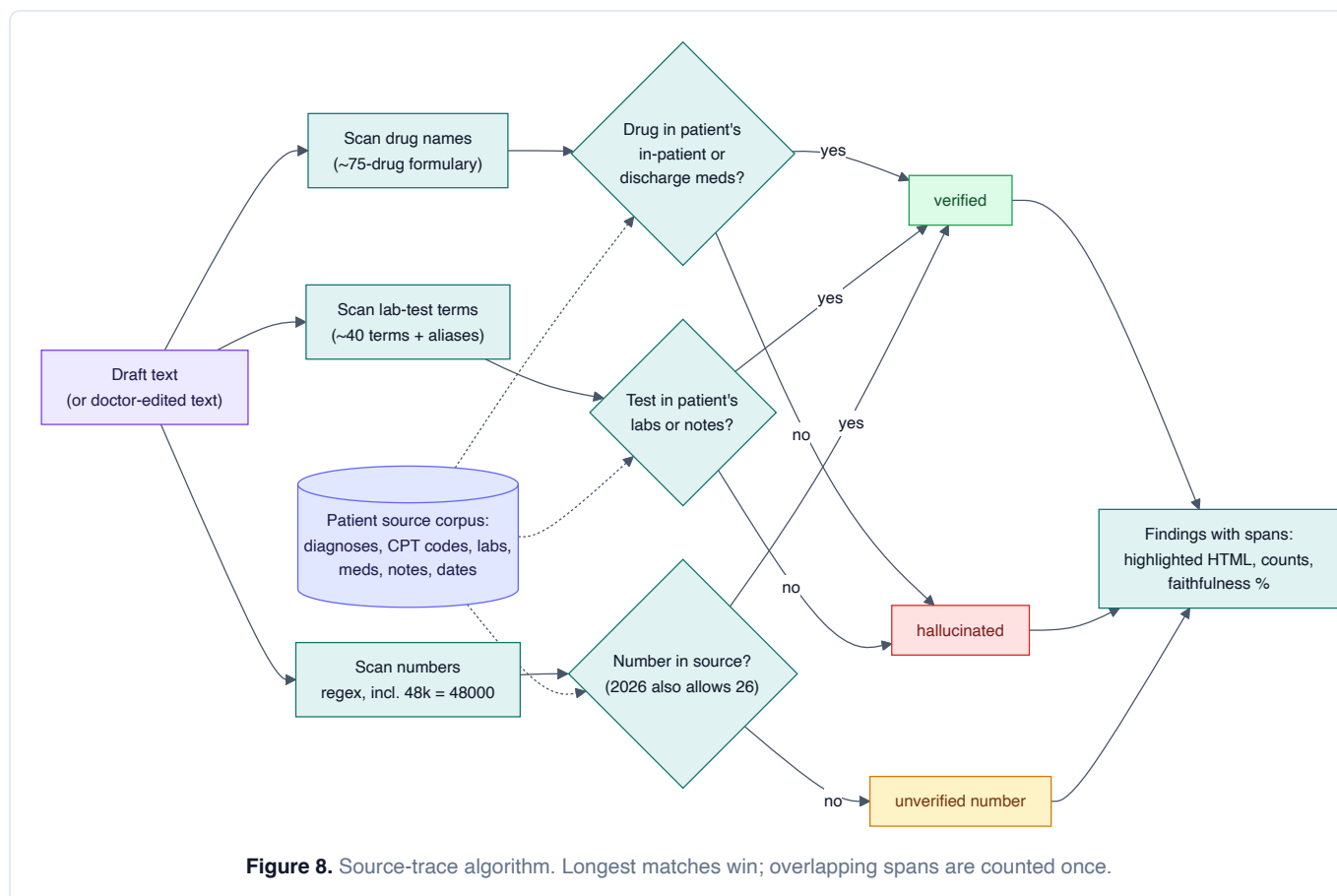
Setting	Value
Model	groq:openai/gpt-oss-20b (env COPILLOT_MODEL)
Output	NativeOutput(SummaryDraft) : JSON schema enforced by the provider; list fields default to empty so partial outputs are not rejected

Setting	Value
Model settings	max_tokens=6000, temperature=0.2, groq_reasoning_effort="low" (caps reasoning so it cannot exhaust the budget)
Retries	Agent validation retries = 2; plus 2 attempts around provider errors (HTTP 400 schema rejections are not retried by the framework)
Pricing (config)	USD 0.075 / 1M input tokens, 0.30 / 1M output tokens; INR at 88/USD

Prompt rules (INSTRUCTIONS)

- Use ONLY facts in the encounter data; never invent drugs, doses, lab values, dates, diagnoses or procedures.
- Copy numbers exactly (values, units, LVEF, stent sizes). Dates as DD/MM/YYYY; never predict a discharge date.
- Do not write a discharge medication list (added from pharmacy). Mention PENDING reports as awaited.
- Hospital course: 4–7 chronological bullets in Indian clinical style (CAG, PTCA, DES, LVEF, TIMI allowed).
- List every procedure. Follow-up / diet only if stated; otherwise "As advised by treating consultant". Fill every schema field.

5.5 Source-trace guardrail



5.6 TPA-readiness rules

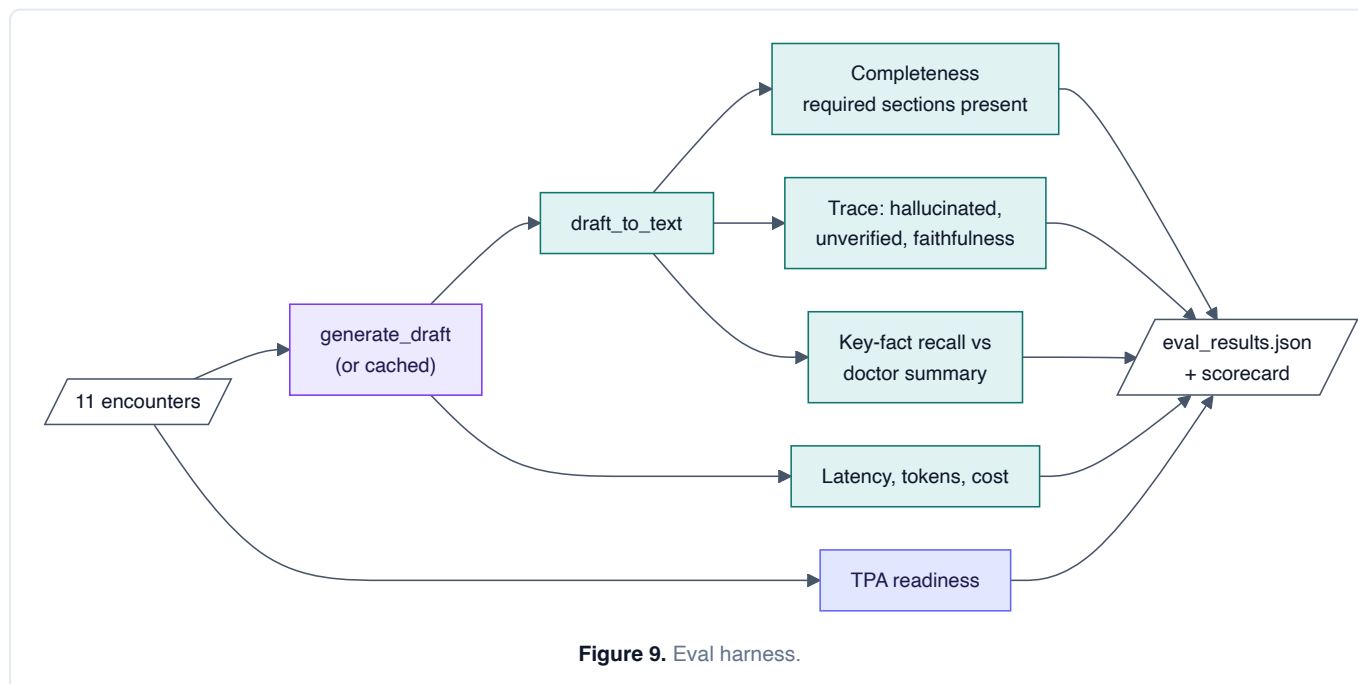
Check id	Rule	Result	Prevents TPA query
final_dx	At least one FINAL diagnosis with ICD-10	pass / fail	Final diagnosis does not match provisional at pre-auth
dx_proc_match	Each procedure's CPT is justified by a diagnosis family (table below)	pass / fail / na	Diagnosis–procedure mismatch

Check id	Rule	Result	Prevents TPA query
required_docs	Documents required by each procedure are in PT_DOC	pass / fail / na	Stent implant sticker not attached; operation notes not attached
pending_reports	PENDING labs are listed in the summary (auto)	warn / pass	Pending report status not mentioned
doctor_reg	Consultant name and registration number present	pass / fail	Treating doctor registration number missing
discharge_meds	Discharge meds exist, each with dose, frequency, duration	pass / warn / fail	n/a (safety)
policy	Cashless: TPA name and policy number present	pass / fail / na	n/a

CPT	Procedure	Justifying ICD-10 families	Required documents
92928	PTCA + stent	I20, I21, I22, I24, I25	CATH_RPT, IMPLANT_STKR
93458	Coronary angiography	I20, I21, I22, I24, I25	CATH_RPT
27447	Total knee arthroplasty	M17	OT_NOTE, IMPLANT_STKR
27125	Hip hemiarthroplasty	S72, M16, M87	OT_NOTE, IMPLANT_STKR
47562	Laparoscopic cholecystectomy	K80, K81, K82	OT_NOTE
44970	Laparoscopic appendectomy	K35, K36, K37	OT_NOTE

Readiness score = passed checks ÷ applicable checks (checks with status *na* excluded). Example: golden patient Ramesh Kulkarni scores 71% (implant sticker missing, one pending report) and 86% once the sticker is uploaded.

5.7 Eval harness



Metric	Definition
Completeness %	Required sections present: presenting complaints, ≥3 hospital-course bullets, key investigations, condition at discharge, follow-up, and procedures (if the patient had any)

Metric	Definition
Hallucinated	Drugs or lab tests mentioned in the draft that are not in the patient's record
Unverified numbers	Numbers not found in the patient's source data
Faithfulness %	Verified entities ÷ all traced entities
Key-fact recall %	Share of drugs, tests and numbers in the doctor's summary (excluding the medication line) that also appear in the draft
Latency, tokens, cost	Per draft, from the agent run usage

5.8 Streamlit UI

Tab	Behaviour
Generate	Encounter picker (sidebar) → source-data expander → Generate → metric strip (source, latency, tokens, ₹ cost, TPA %) → assembled summary with  sections + TPA checklist
Source trace	Editable draft text re-traced on every edit; colour legend; highlighted text; table of non-verified findings (add "Warfarin 5 mg OD" to demo)
Eval	Run all encounters (optionally reuse cache) with progress bar; KPIs + per-encounter table; persisted to poc/data/eval_results.json
Clinician feedback	AI draft vs doctor original side-by-side; reviewer, three 1–5 ratings, "would sign?", comments → feedback.jsonl; aggregate KPIs

Drafts are kept in `st.session_state["drafts"]` and pre-loaded from the last eval run, so every tab works without a new LLM call.

5.9 De-identified export

`poc/export_deidentified.py` opens MediTrack read-only (`mode=ro`), selects discharged admissions that have a typed summary, builds each `Encounter`, then applies `deidentify()`: IP No → `ENC-nn`, name → `Patient ENC-nn`, UHID → SHA-256 prefix, policy → XXXX. Clinical content is preserved.

5.10 Results (live model, 11 encounters)

0 Hallucinated drugs / tests	99% Faithfulness	74% Key-fact recall	87% Completeness
0.6 s Median latency	\$0.0002 Cost per summary (~₹0.02)	19 / 19 Live drafts after fixes	58 / 58 Golden draft entities verified

Finding	Resolution
Groq rejected gpt-oss tool-call output; all drafts silently fell back to template	Switched to <code>NativeOutput</code> . Lesson: test with the real provider, not just the framework.
Some drafts failed schema validation (missing list fields) or exhausted tokens reasoning	Optional list fields, reasoning effort "low", max_tokens 6000, provider-error retry: 16/19 → 19/19 live
ENC-01: model wrote wrong dates	Caught by guardrail as unverified numbers; doctor corrects before signing

Finding	Resolution
Follow-up often missing	Not a model bug: the follow-up plan is not written in notes. Process fix: a follow-up line in the final ward note.

6. Integrated system: High-Level Design

6.1 Design principles

Principle	How it is realised
Do not touch the legacy core	Read only from the MIS replica; write back only through MediTrack's own import hot-folder; entry point via configuration
Human in the loop	AI drafts; the treating doctor edits and signs. Nothing is sent without a signature.
Adoption over novelty	Doctors never leave MediTrack: the review screen is a patient-file tab in MediTrack's classic look
Deterministic where possible	Diagnoses, meds, pending reports and TPA rules come from data and code, not the LLM
Always available	Fallback chain (live → cached → template); doctors can always use the old Word template
Observable and auditable	Logfire traces for every step; audit events in the Copilot store

6.2 Architecture

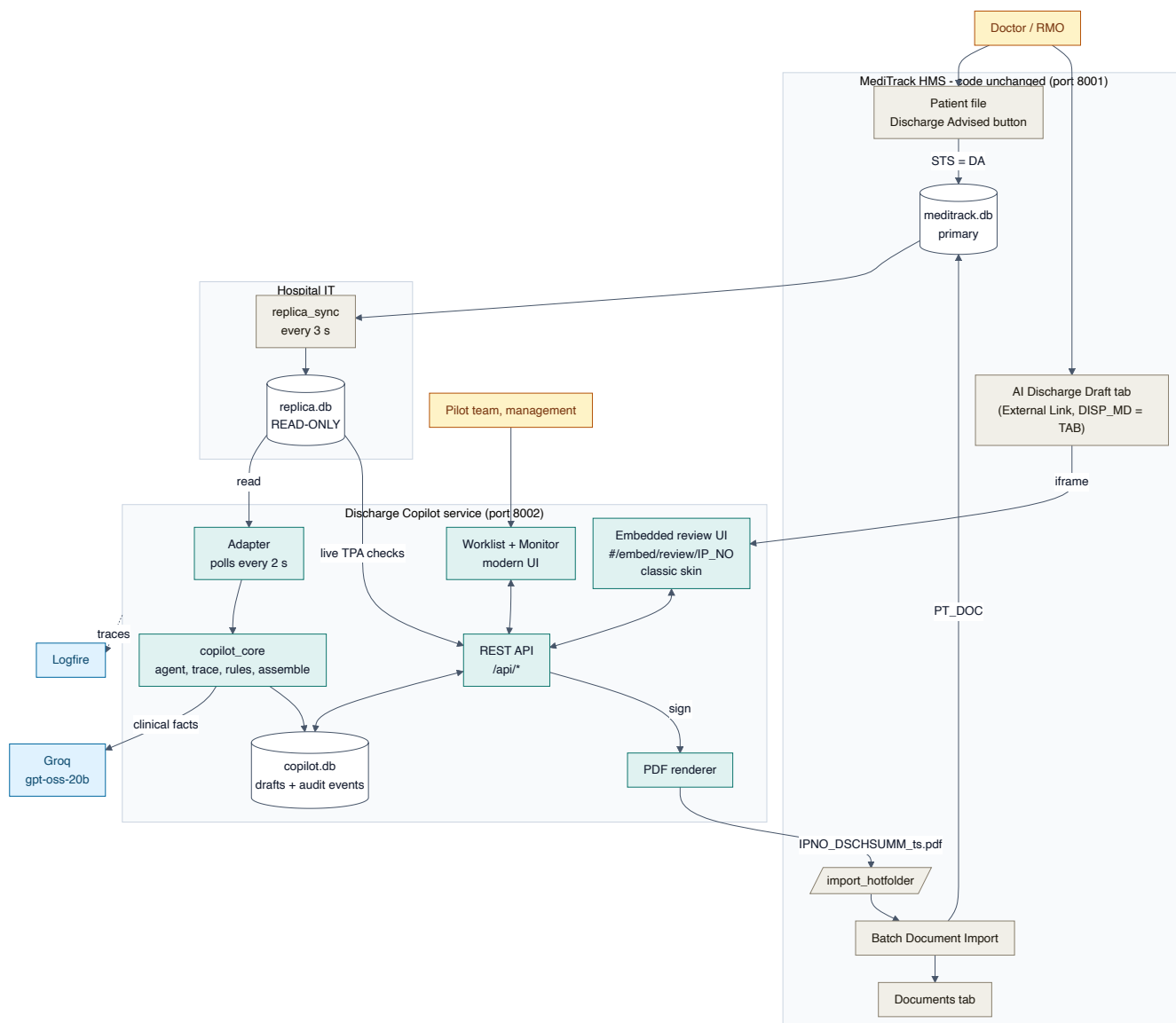


Figure 10. Integrated architecture. The Copilot touches MediTrack only through three pre-existing doors: replica (read), External Link tab (entry), hot-folder (write).

6.3 End-to-end flow

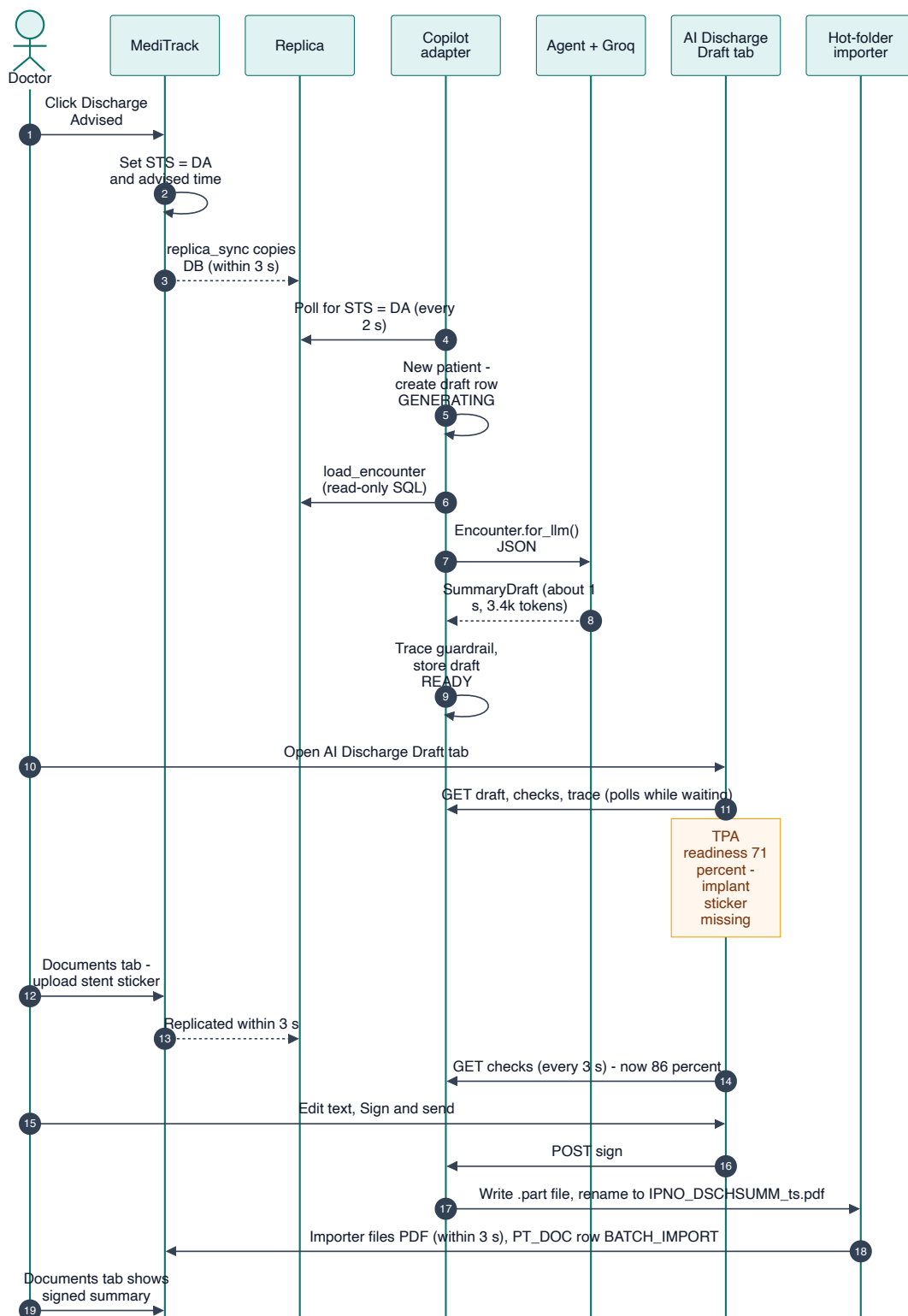


Figure 11. Golden-path sequence for patient IP2609-0142. Draft is ready about 5–7 s after "Discharge Advised".

6.4 Deployment view

Process	Pilot (demo)	Production equivalent
MediTrack	uvicorn :8001 (FastAPI + SQLite)	Existing MediTrack Java/Oracle servers (unchanged)

Process	Pilot (demo)	Production equivalent
Replica	<code>copilot.replica_sync</code> (SQLite backup API, atomic swap)	Oracle Data Guard / MIS reporting replica owned by IT
Copilot service	<code>uvicorn :8002</code> (API + adapter thread + built React UI)	Container in hospital VPC / on-prem VM; health-checked
Copilot store	SQLite <code>copilot/data/copilot.db</code>	PostgreSQL with backups
LLM	Groq API	Zero-data-retention endpoint or approved in-region host
Observability	Logfire (if token set)	Logfire / hospital OpenTelemetry collector
Launch	<code>./run.sh</code> integrated (starts all three)	systemd / Kubernetes

6.5 Security & compliance

Concern	Control
Legacy integrity	No writes to MediTrack DB; replica opened with <code>mode=ro</code> ; write-back via MediTrack's own importer
Data minimisation (DPDP)	<code>Encounter.for_llm()</code> strips name, UHID, policy number, TPA, registration number, documents, discharge meds
Clinical safety	Deterministic  sections; guardrail; mandatory doctor signature; sign dialog warns about failing TPA checks
Audit	<code>event</code> table: detection, draft ready, edits (% changed), signature with signer and file name
Secrets	API keys only in <code>.env</code> (git-ignored)
Observability data	Logfire scrubbing disabled for synthetic demo data only; enabled in production

6.6 Non-functional requirements

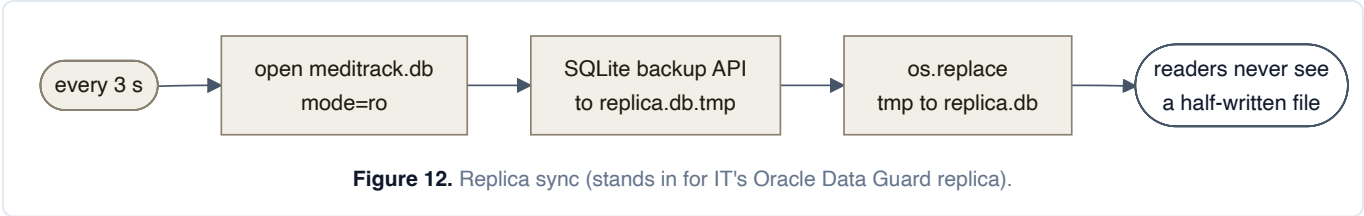
Requirement	Target	Measured in pilot
Draft ready after "Discharge Advised"	< 60 s	~5–7 s (replica ≤3 s + poll ≤2 s + LLM ~1 s)
TPA check refresh after document upload	< 10 s	≤6 s (replica 3 s + UI poll 3 s)
Signed PDF visible in MediTrack	< 10 s	≤3 s (importer interval)
Availability when LLM is down	Always a draft	Cached → template fallback
Cost per summary	< ₹1	~₹0.02

7. Integrated system: Low-Level Design

7.1 Components

Component (file)	Responsibility
copilot/replica_sync.py	Copies meditrack.db → replica.db.tmp via SQLite backup API, then os.replace (atomic) every 3 s
copilot/service/app.py	FastAPI app: adapter thread, REST API, metrics, static UI
copilot/service/store.py	Copilot's own SQLite store (draft , event), read-only replica connection
copilot/service/pdf.py	Signed summary PDF (fpdf2), Latin-1 safe
copilot/ui	React SPA: #/embed/review/:ip (classic skin, in MediTrack), #/review/:ip , #/worklist, #/monitor
copilot_core/*	Shared with the POC (Section 5)

7.2 Replica sync



7.3 Adapter & draft lifecycle

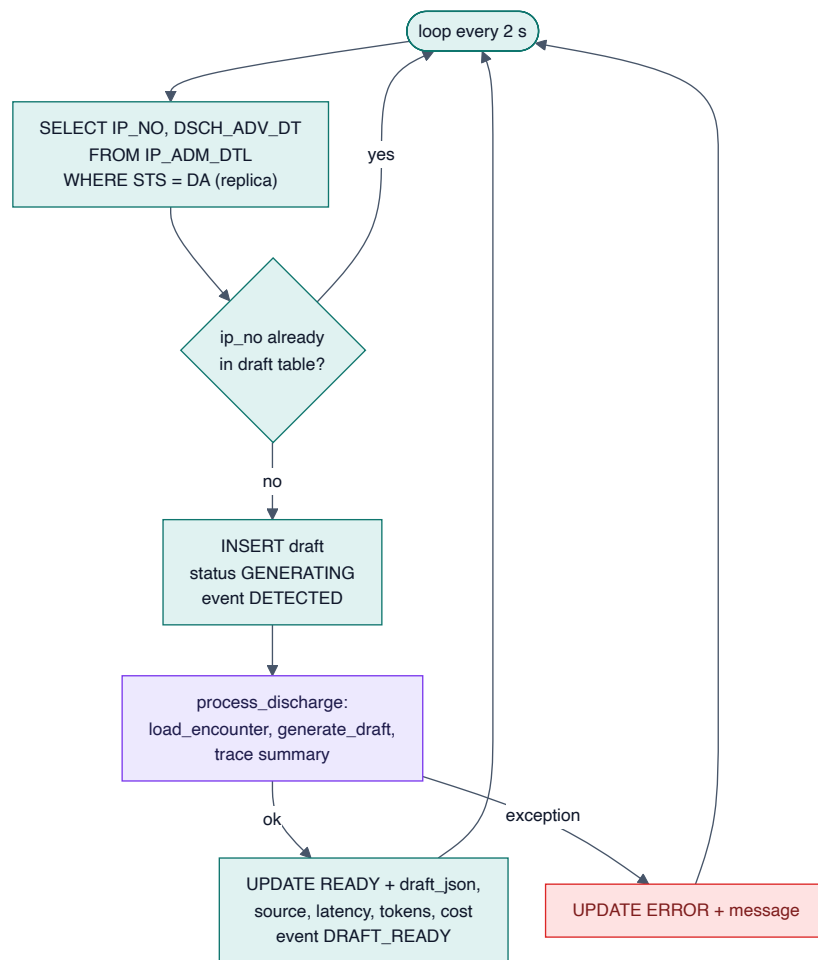


Figure 13. Adapter loop.

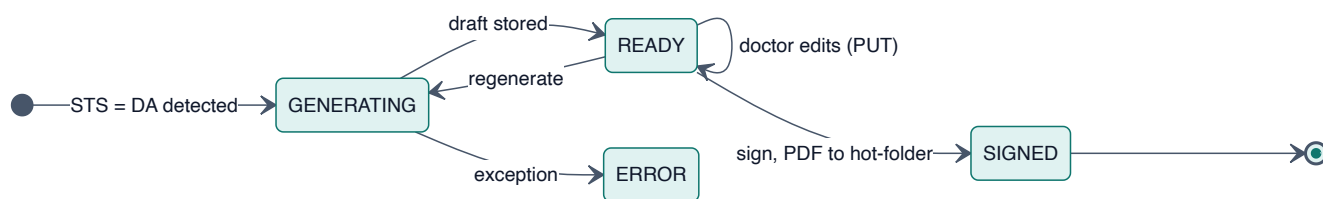


Figure 14. Draft lifecycle.

7.4 Copilot datastore

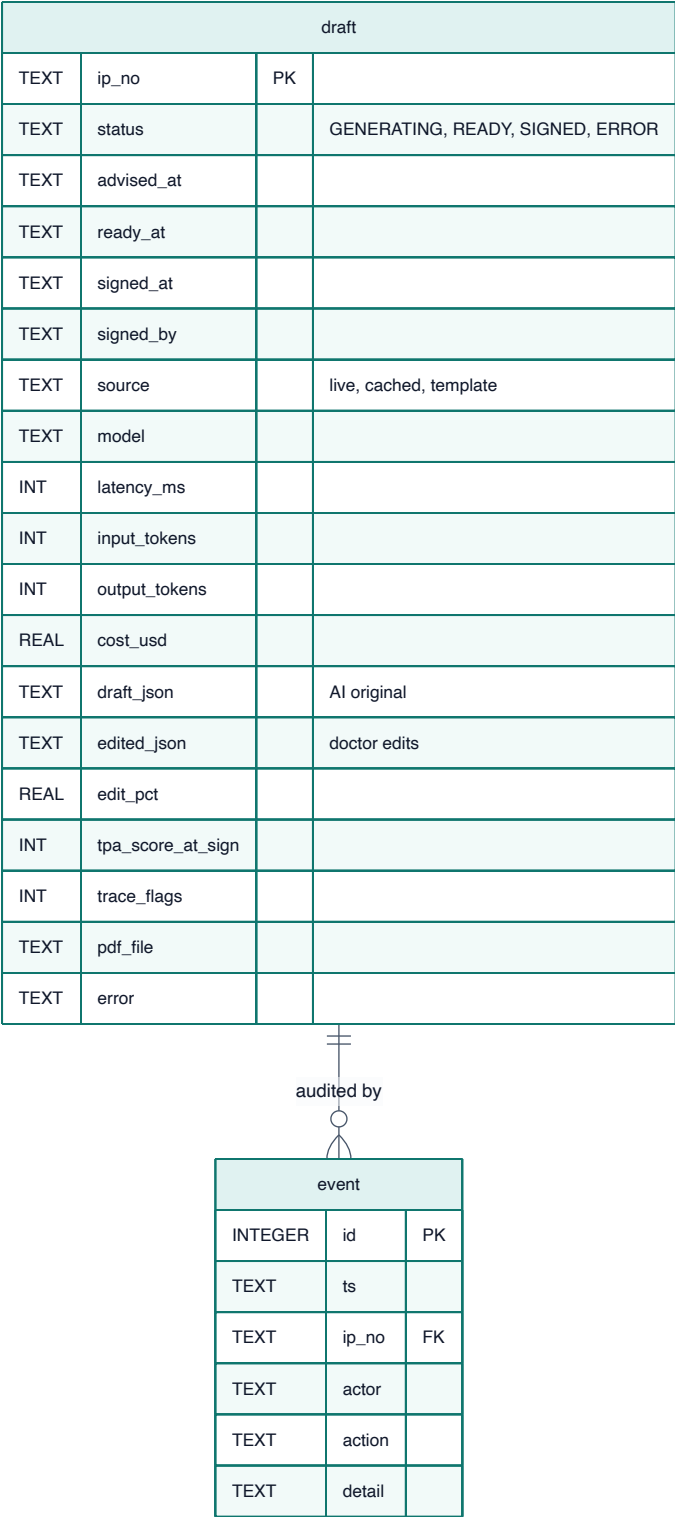


Figure 15. Copilot store. The AI original and the doctor's edits are both kept, which gives an edit-distance metric and a full audit trail.

7.5 REST API

Method	Path	Behaviour
GET	/api/drafts	Worklist: drafts joined with patient, ward, consultant, payment from the replica
GET	/api/drafts/{ip_no}	Meta, fresh Encounter (replica), current draft (edited if any), TPA checks + readiness, trace summary + flags. 404 until discharge advised.

Method	Path	Behaviour
GET	/api/drafts/{ip_no}/checks	TPA checks re-evaluated from the replica on every call
PUT	/api/drafts/{ip_no}	Save doctor edits ({draft: SummaryDraft}); computes edit % with diffliab; returns new trace. 409 if signed.
POST	/api/drafts/{ip_no}/regenerate	Discard edits, re-run the pipeline
POST	/api/drafts/{ip_no}/sign	{signed_by} → render PDF → hot-folder → status SIGNED, readiness at sign stored
GET	/api/drafts/{ip_no}/pdf	Unsigned PDF preview
GET	/api/metrics	Baseline (replica) + pilot metrics + last 25 events + Logfire URL

7.6 Embedded review UI (the AI Discharge Draft tab)

Aspect	Design
Entry	MediTrack External Link: label AI Discharge Draft , URL <code>http://localhost:8002/#/embed/review/{IP_NO}</code> , display Patient file tab → rendered in an iframe inside the patient file
Look	<code>.classic</code> skin: Tahoma 11px, grey #ece9d8 panels, groove borders, bevelled buttons, so it looks native and needs minimal training
Before discharge advised	Waiting message; polls every 2 s until the draft exists
Editing	AI sections as textareas (lists: one item per line; investigations "test: result (interpretation)");  diagnoses, pending reports, discharge meds are read-only
Live panels	TPA readiness re-pollled every 3 s (flashes on change); source-trace counts and flags
Actions	Save edits · Regenerate · Preview PDF · Sign & send to MediTrack
MediTrack side	Every tab click refreshes patient data (classic round-trip), so the signed PDF appears in Documents without a page reload

7.7 Sign & write-back

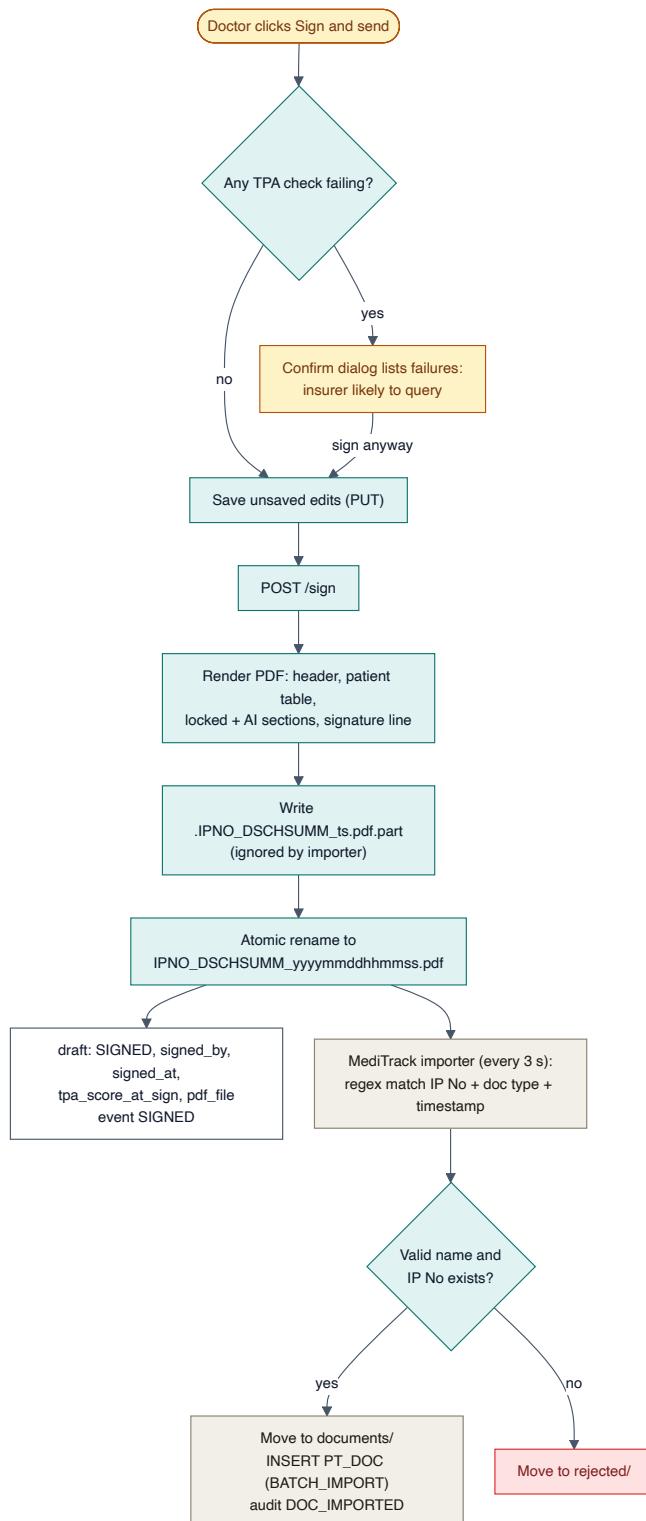


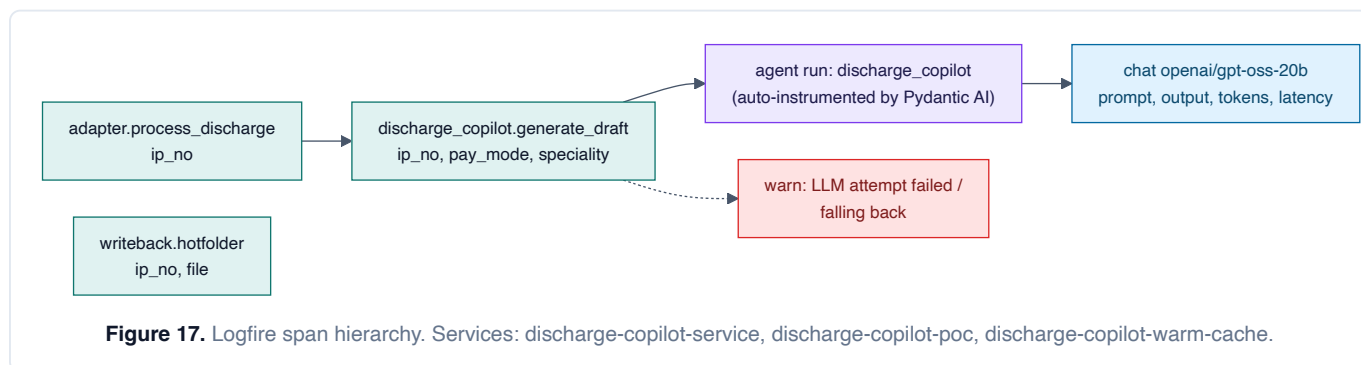
Figure 16. Sign and write-back through MediTrack's own importer. Filename regex: `^(IP\d{4}-\d{4})_([A-Z_]+?)_(\d{14})\.pdf$`

7.8 Metrics

Metric	Computation
Baseline TAT (cash / cashless)	AVG(DSCH_DT – DSCH_ADV_DT) in hours from the replica, by PAY_MODE
Baseline TPA query %	QUERY ÷ all rows in TPA_CLM_DTL

Metric	Computation
Draft ready (s)	Median LLM latency of drafted rows
Advised → signed (min)	Median(signed_at – advised_at)
TPA readiness at sign	Mean of tpa_score_at_sign
Doctor edit %	Mean of edit_pct (1 – difflib ratio of AI vs edited text)
Guardrail flags · AI cost · sources	Sum of trace flags · sum of cost_usd · count by live / cached / template

7.9 Observability



7.10 Failure modes

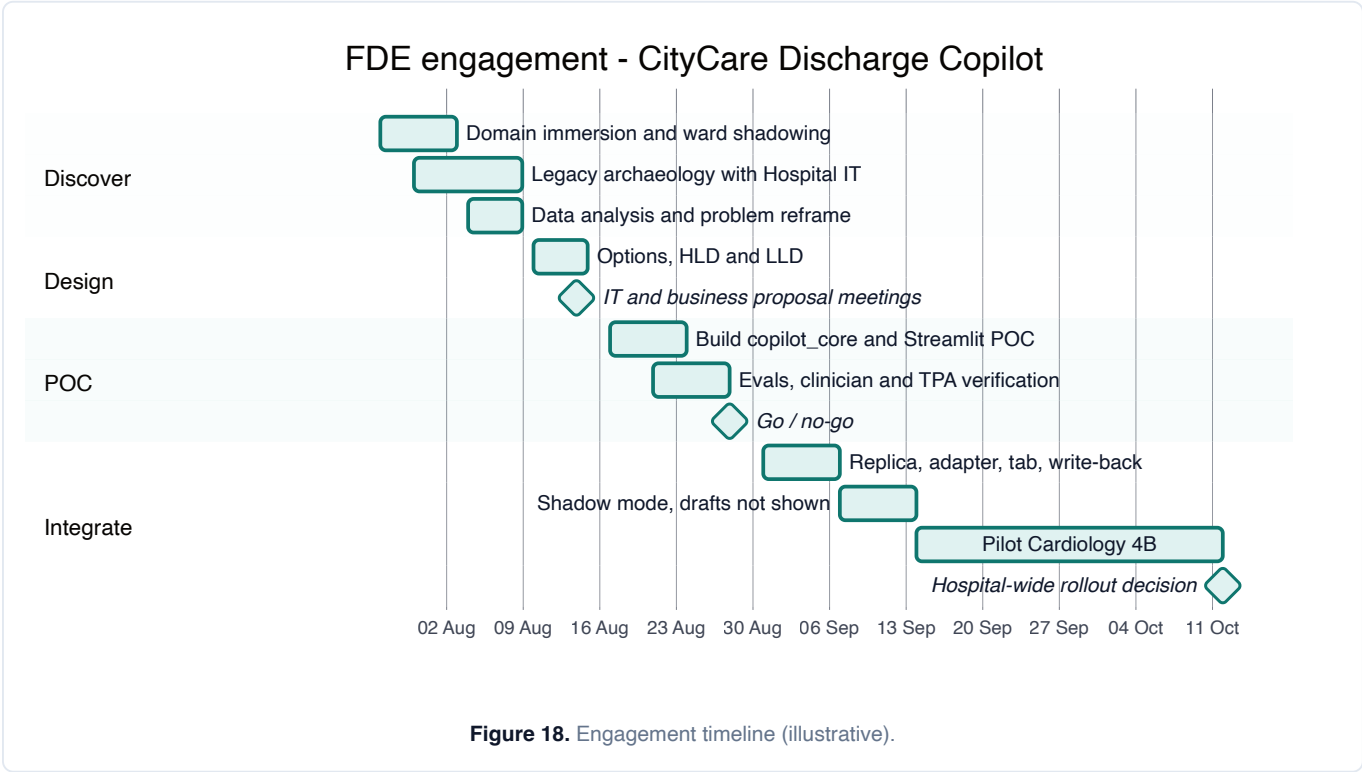
Failure	Behaviour	Recovery
Groq down / network loss	Cached draft served (source = cached) or template	Automatic; <code>./run.sh warm</code> keeps cache fresh
Schema rejection (HTTP 400)	One retry, then fallback	Logged as Logfire warning
Replica stale / missing	Adapter waits; API returns 404 for unknown patients	Sync recreates replica within 3 s
Pipeline exception	Draft status ERROR with message	Regenerate from UI
Hot-folder file invalid	MediTrack moves it to <code>rejected/</code>	Filename convention enforced by Copilot
Partial PDF picked up	Prevented: <code>.part</code> temp name + atomic rename	n/a
Copilot service down	Tab shows an error; MediTrack is unaffected	Doctors use the Word template as before

7.11 Configuration

Setting	Purpose
GROQ_API_KEY, LOGFIRE_TOKEN	LLM access; tracing (optional)
COPILOT_MODEL	Default <code>groq:openai/gpt-oss-20b</code>
PRICE_INPUT_PER_M, PRICE_OUTPUT_PER_M, USD_INR	Cost reporting
COPILOT_OFFLINE=1	Never call the LLM (cache / template only)
LOGFIRE_PROJECT_URL	Link from the Monitor page
MediTrack SYS_EXT_LNK row	AI Discharge Draft · <code>http://localhost:8002/#/embed/review/{IP_N0}</code> · TAB

Setting	Purpose
Intervals	Replica 3 s · adapter 2 s · importer 3 s · UI checks 3 s · worklist / waiting 2 s

8. Rollout plan



Phase	Exit criteria
Shadow mode	Drafts generated for every discharge but hidden; 0 hallucinations; readiness scores agree with TPA outcomes
Pilot (Ward 4B)	>80% of discharges signed via the tab; cashless TAT <3.5 h; TPA query rate <15%; 20-minute training delivered
Expansion	Hospital-wide rollout; next use case: pre-authorization at admission (same Encounter + rules pattern)

Appendix A: Repository map & run commands

```
Live_session/
├─ legacy_meditrack/          Part 1 – MediTrack HMS v3.2
│   ├── backend/ app.py db.py schema.sql seed.py seed_data.py
│   └── ui/      React (classic 2011 look)
├─ copilot_core/             Part 2 – AI core (shared)
│   ├── models.py encounter.py agent.py trace.py rules.py assemble.py
│   ├── observability.py config.py
│   └── cache/      warm-cache fallback drafts
├─ poc/                      Part 2 – Streamlit POC
│   ├── app.py evaluation.py export_deidentified.py
│   └── data/deidentified_encounters.json eval_results.json
├─ copilot/                  Part 3 – Integrated service
│   ├── replica_sync.py
│   ├── service/ app.py store.py pdf.py
│   └── ui/      React (worklist, monitor, embedded classic review)
├─ scripts/warm_cache.py
├─ demo_assets/IP2609-0142_stent_implant_sticker.pdf
├─ webinar/ webinar-content.md system-explainer.md
├─ docs/design/ hld_lld.html (this document's source)
└─ run.sh  README.md DEMO_RUNBOOK.md .env.example
```

Command	Effect
./run.sh setup	Create venv, install Python and npm dependencies, seed data
./run.sh legacy	MediTrack only → http://localhost:8001
./run.sh poc	Streamlit POC → http://localhost:8501
./run.sh integrated	MediTrack :8001 + replica sync + Copilot :8002
./run.sh warm	Generate and cache live drafts for all patients
./run.sh reset	Reseed MediTrack and clear Copilot state (golden demo state)